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Unit – 1 : Session – 4 : SLO – 2 | Implementation of Structures Using Pointers — Fill in the Blanks

■ I. Fill in the Blanks

1. * (asterisk) — a structure pointer is declared using the * operator: struct S *ptr;
2. -> (arrow) — the arrow operator is used to access members via a pointer.
3. the address — a structure pointer stores the address of a structure variable.
4. &var — ptr = &var; assigns the address.
5. direct (efficient member) — pointers enable direct access to the structure's members without copying.
6. allocated (initialized) — memory for the pointed structure must be allocated before use.
7. in-place (permanent) — passing a pointer to a function lets it modify the original structure.
8. ptr->member — (*ptr).member is the same as ptr->member.
9. structure (data) — pointers reduce structure copying overhead.
10. malloc — dynamic allocation: ptr = malloc(sizeof(struct S));
11. size (amount of memory) — sizeof(struct) ensures the correct allocation size.
12. freed — unused structure pointers' memory should be freed.
13. dynamic — pointers enable dynamic structures like linked lists.
14. dangling pointers (undefined behavior) — always initialize to NULL or a valid address.
15. member access (the dot) — the arrow operator combines dereference and member access.